Term-1 21/11/23 M/S

USN. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [23BSPH011]

FIRST SEMESTER B. Tech EXAMINATION, NOVEMBER - 2023

ENGINEERING PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:

i. Missing data may be suitably assumed.

ii. Physical constants

(i) Mass of the electron $m_e = 9.1 \times 10^{-31}$kg

(ii) Mass of proton, neutron $m_n = m_p = 1.67 \times 10^{-27}$kg

(iii) Charge of the electron $e = 1.602 \times 10^{-19}$C

(iv) Planck's constant $h = 6.626 \times 10^{-34}$Js

(v) Boltzmann's constant $k = 1.38 \times 10^{-23}$J/K

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a**	Explain the concept of matter waves	L2	CO1	**02**
	b.	Define angle of acceptance for an optical fiber.	L1	CO2	**02**
	c.	Explain effective mass of Electron.	L2	CO3	**02**
	d.	What is a Mach number?	L1	CO4	**02**
	e.	State and Explain Hooke's law.	L2	CO5	**02**
		PART-B	**6X5=30**		
2.	**a.**	What is de-Broglie wavelength. For an electron accelerated by the applied potential, arrive at the expression for de-Broglie wavelength.	L2	CO1	**05**
		OR			
	b.	Explain the Physical Significance of a Wave function.	L2	CO1	**05**
3.	**a.**	Outline the characteristic properties of Matter waves.	L1	CO1	**05**
		OR			
	b.	Calculate the first 3 permitted energy values for an electron in a box of width 4Å.	L3	CO1	**05**
4.	**a.**	Explain the basic interactions of Radiation with matter.	L2	CO2	**05**
		OR			
	b.	With neat labelled diagram explain any two types of optical fibers.	L2	CO2	**05**
5.	**a.**	Review the assumptions of classical free electron theory.	L3	CO3	**05**
		OR			
	b.	Explain Forward and reverse bias characteristics of a Zener diode.	L2	CO4	**05**

JGU (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

6.	**a.**	Derive the differential equation for simple harmonic motion.	L2	CO5	**05**
		OR			
	b.	What are shock waves. Explain the properties of Shock Waves.	L2	CO5	**05**
7.	**a.**	Explain the different types of Modulus of Elasticity.	L2	CO6	**05**
		OR			
	b.	What is Bending moment of a beam. Write its applications.	L3	CO6	**05**
		PART-C	**6X10=60**		
8.	**a.**	**i.** Describe Davisson and Germer`s experiment for confirmation of De-Broglie hypothesis.	L2	CO1	**07**
		ii. A particle of mass 0.5MeV $/c^2$ has kinetic energy 100eV. Find its de-Broglie wavelength, where c is the velocity of light	L3	CO1	**03**
		OR			
	b.	**i.** State Heisenberg`s uncertainty principle and Prove the Non-existence of an electron inside the Nucleus using it.	L3	CO1	**06**
		ii. An electron has a speed of 100m/s. The inherent uncertainty in its measurement is 0.005%. Calculate the corresponding uncertainty that arises in the measurement of its position.	L3	CO1	**04**
9.	**a.**	**i.** Set up time independent one-dimensional Schrodinger's wave equation.	L2	CO1	**07**
		ii. The velocity of an electron was measured to be 5×10^5 m/s with an uncertainty of 1%. What is the uncertainty involved in the measurement of position?	L3	CO1	**03**
		OR			
	b.	Assuming the time independent Schrodinger wave equation obtain the normalized wave function and discuss the solutions for energy of a particle in one dimensional potential well of infinite height.	L3	CO1	**10**
10.	**a.**	Calculate the wavelength of laser beam using the following data. Grating constant C = 5.08×10^{-5} ; f = 100 cm	L2	CO2	**10**

Diffraction order (n)	Average distance (x in cm)
1	1.1
2	2.2
3	3.3
4	4.4
5	5.6

OR

	b.	i. Obtain an expression for numerical aperture in terms of refractive indices of core and Cladding and then arrive at the condition for propagation.	L2	CO2	**07**
		ii. Find the numerical aperture and angle of acceptance of an optical fiber, if the refractive index of core and cladding are 1.5 and 1.48 respectively.	L3	CO2	**03**
11.	**a.**	i. What is Fermi energy? Explain the Variation of Fermi factor with temperature.	L2	CO3	**06**
		ii. Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.	L3	CO3	**04**
		OR			
	b.	i. With neat diagram, Review the Band theory of Solids.	L2	CO4	**05**
		ii. Differentiate between intrinsic and extrinsic semiconductors.	L4	CO4	**05**
12.	**a.**	Give the theory of Damped oscillations and obtain the relation for decay constant.	L2	CO5	**10**
		OR			
	b.	i. Explain the construction and working principle of Reddy`s shock tube.	L2	CO5	**06**
		ii. Mention the applications of Reddy`s shock tube.	L3	CO5	**04**
13.	**a.**	Obtain the relation between Young's modulus, rigidity modulus and Poisson's ratio when the force is applied on one face of a cube.	L-3	CO6	**10**
		OR			
	b.	Obtain an equation for the Young's modulus when a downward force is applied to beam of a single cantilever. (Use the relation for the bending moment $= \frac{Y}{R} I_g$).	L-3	CO6	**10**